

Department of Computer Science & Engineering

Subject: Computer Network

Subject Code: BCS502

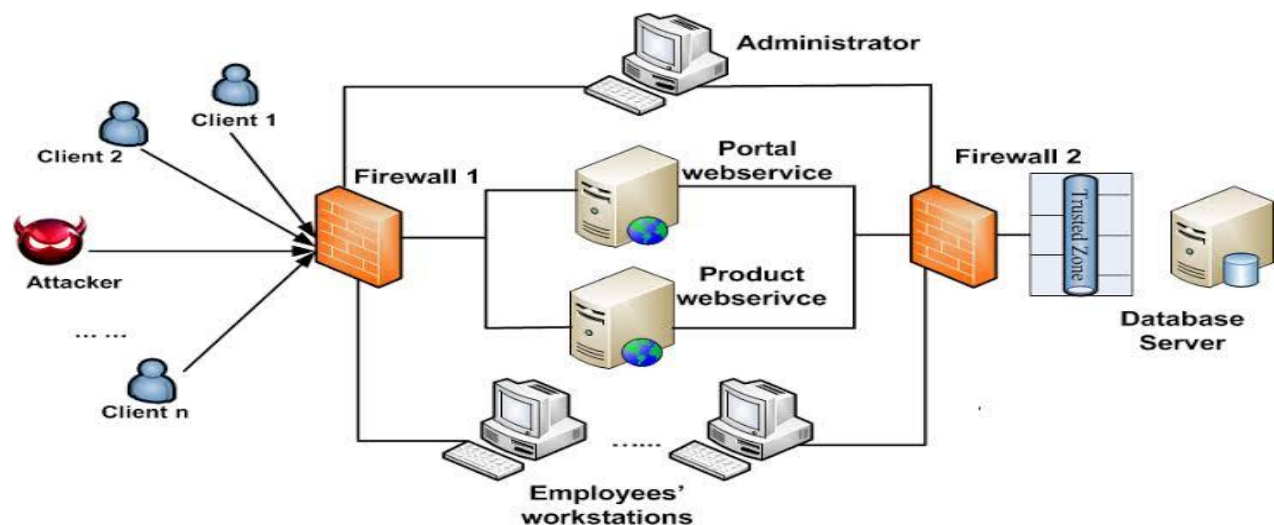
Group :

Topic: “Network Forensic Techniques”

Introduction:

Network forensic techniques are used to find and study evidence of cybercrimes that happen over a computer network. They help experts track hackers, detect security attacks, and understand how the attack took place. These techniques include checking network traffic, analyzing logs, and using special tools to monitor data. In short, network forensics helps protect systems, find the source of problems, and keep digital information safe.

Architecture:



Advantages of Network Forensic Techniques:

- Helps detect and investigate cyberattacks quickly.
- Provides digital evidence useful for legal and security purposes.
- Improves network security by identifying vulnerabilities.
- Monitors real-time traffic to prevent future attacks.

Disadvantages of Network Forensic Techniques:

- Requires expensive tools and skilled professionals.
- Analyzing large amounts of network data can be time-consuming.
- May raise privacy concerns due to constant monitoring.
- Difficult to trace attacks if data is encrypted or deleted.

Applications of Network Forensic Techniques:

- Used by cybersecurity teams to detect and analyze network attacks.
- Helps law enforcement collect digital evidence for cybercrime cases.
- Supports organizations in monitoring employee or system activities.
- Assists in identifying data breaches and preventing information leaks.

Future Scope of Network Forensic Techniques:

- Integration with **AI and Machine Learning** for faster and smarter threat detection.
- Use of **Cloud and IoT forensics** to handle modern connected environments.
- Adoption of **Blockchain technology** for secure evidence storage.

Group picture:



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